StealthRL: Reinforcement Learning Paraphrase Attacks for Multi-Detector Evasion of AI-Text Detectors

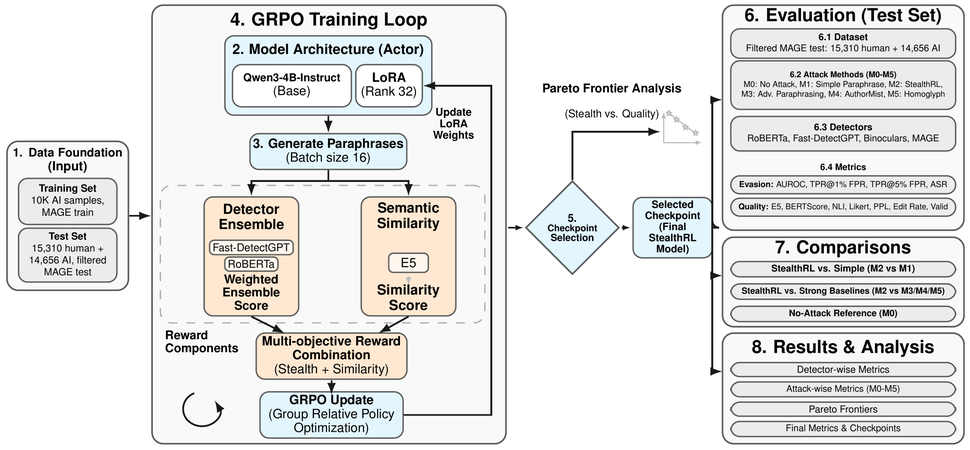
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Code Available

0.024 mean TPR@1%FPR | **97.6%** mean ASR | mean AUROC 0.79 → 0.43
4 detectors: RoBERTa, Fast-DetectGPT, Binoculars, MAGE • **Transfers to 2 held-out detectors**

Method



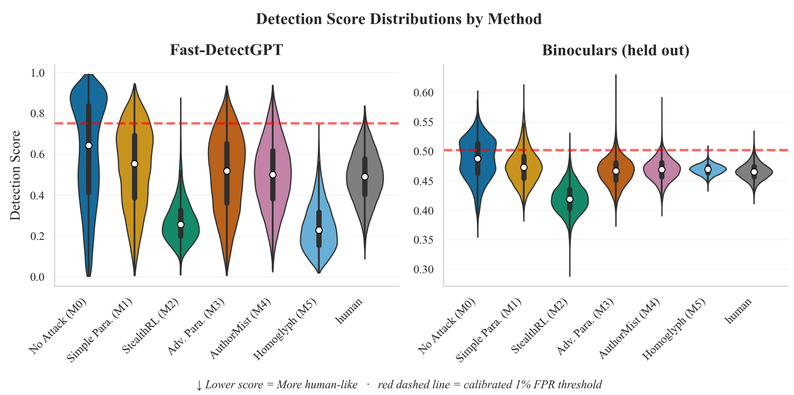
Reward: $R = \alpha(1 - p_{ens}) + \beta \cdot sim_{E5}$, $\alpha = 1.0, \beta = 0.1$

Detector Ensemble: $p_{ens} = 0.6 \cdot p_{RoBERTa} + 0.4 \cdot p_{Fast\text{-}DetectGPT}$

Base Model: Qwen3-4B-Instruct + LoRA (rank 32)

- **RL Algorithm:** GRPO, 3 epochs, 10K AI-generated MAGE samples
- **Threat Model:** Black-box (score-only, no gradients)
- **Transfer Test:** Binoculars and MAGE held out during training

Why Evasion Works: Score Distributions



↓ Lower score = More human-like · red dashed line = calibrated 1% FPR threshold

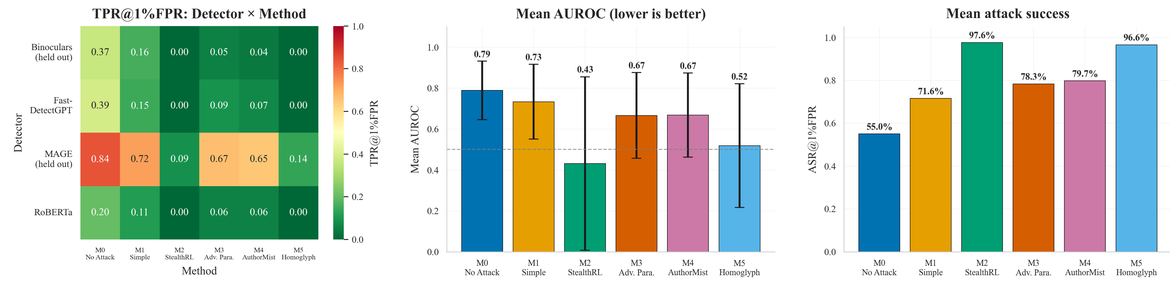
StealthRL (M2) pushes most AI-sample scores below the calibrated 1% FPR threshold, so TPR collapses even where a detector such as RoBERTa keeps some global rank information. Binoculars (held out) AUROC drops from 0.705 to 0.055.

Experimental Results

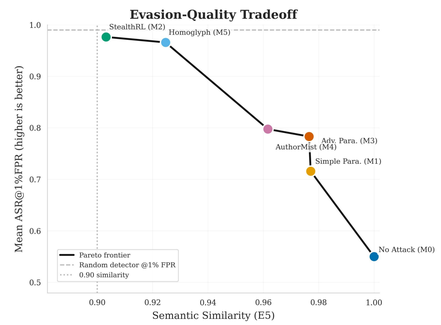
TPR@1%FPR: 6 attack methods × 4 detectors, MAGE test pool

Method	RoBERTa TPR@1%	Fast-DetGPT TPR@1%	Binoculars TPR@1% (held-out)	MAGE TPR@1% (held-out)	Mean TPR↓	Mean ASR↑
M0 No Attack	0.203	0.388	0.367	0.843	0.450	0.550
M1 Simple Para.	0.111	0.148	0.158	0.720	0.284	0.716
M2 StealthRL	**0.002**	**0.002**	**0.002**	**0.089**	**0.024**	**0.976**
M3 Adv. Para.	0.058	0.092	0.051	0.666	0.217	0.783
M4 AuthorMist	0.056	0.067	0.036	0.651	0.203	0.797
M5 Homoglyph	0.001	0.000	0.000	0.136	0.034	0.966

TPR@1%FPR heatmap, mean AUROC, and mean attack success



Evasion–quality tradeoff



StealthRL Quality

0.901 E5 Similarity	**0.868** BERTScore
2.51/5 Likert Quality	**0.316** NLI Consistency

Key Findings

- **Robustness breakdown at strict operating point:** StealthRL cuts mean TPR@1%FPR to 0.024 across four detectors (97.6% ASR), using one generation per input.
- **Transfer to held-out detectors:** On Binoculars and MAGE (never used in training), TPR@1%FPR falls from 0.367 to 0.002 and from 0.843 to 0.089.
- **Stable across runs and operating points:** The original run and two re-seeded repeats give mean TPR@1%FPR 0.024 ± 0.001; at 5% FPR, mean TPR is 0.076.
- **Evasion–fidelity tradeoff:** More readable than homoglyphs (Likert 2.51 vs 1.97), but less faithful than M1/M3 (NLI 0.316 vs 0.967/0.965).